

DIGITAL LOGIC and DESIGN

Lesson 2- Brief History of Number System

Brief History of Number System

Intended Learning Outcome (ILOs):

After completing this chapter, students will be able to:

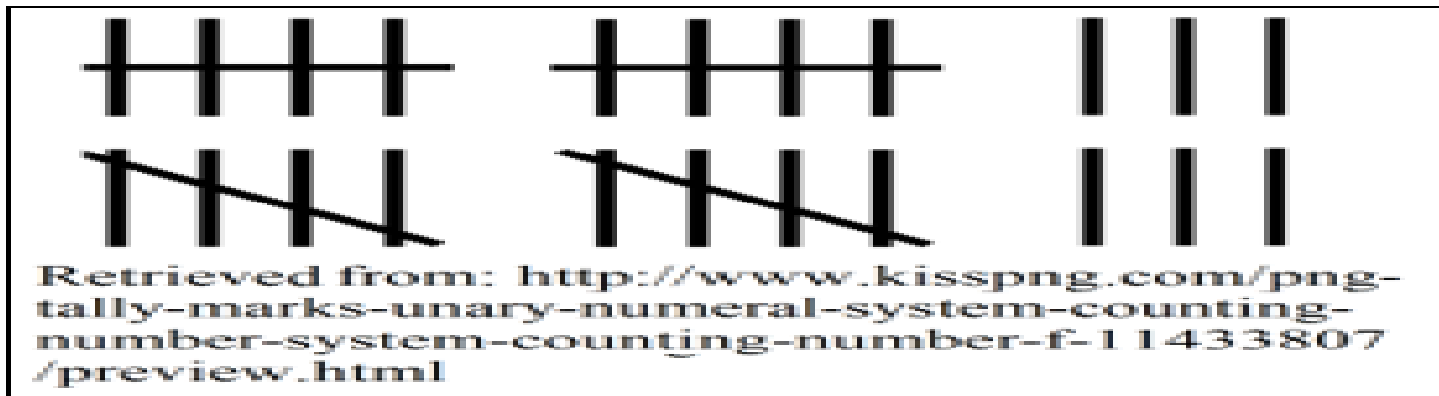
- understand the history of number system.
- identify the different number system.
- recognize the generation of computer in relation with number system
- appreciate Moore's Law
- comprehend the scale of integration of integrated circuit

Brief History of Number System

2.1 BRIEF HISTORY OF NUMBER SYSTEM

2.1.1 Unary System

- The unary numeral system is the bijective base-1 numeral system. It is the easiest numeral system to express normal numbers. In to express a number N , an arbitrary chosen symbols to represent 1 is repeated N times.



Brief History of Number System

- These numbers ought to be recognized from reunits, which are additionally composed as groupings of ones yet have their standard decimal numerical understanding. This system is utilized in counting. For instance, utilizing the count check |, the number 3 is spoken to as | | |. In East Asian societies, the number three is spoken to as "三" (1 and 2 are spoken to a similar way), a character that is drawn with three strokes.

Brief History of Number System

2.1.2 Roman Numeral

- Roman numeral, any of the symbols or images utilized in an arrangement of numerical representation dependent on the old Roman system. The symbols are I, V, X, L, C, D, and M, standing independently for 1, 5, 10, 50, 100, 500, and 1,000 in the Hindu-Arabic numeral system. An image placed after another of equivalent or more remarkable value includes its value; e.g., II = 2 and LX = 60. An image set before one of more prominent value subtracts its value; e.g., IV = 4, XL = 40, and CD = 400. A bar put over a number increases its value by 1,000.

Brief History of Number System

ROMAN NUMERALS IN ENGLISH

I	One	XI	Eleven	XXX	Thirty	CD	Four hundred
II	Two	XII	Twelve	XL	Forty	D	Five hundred
III	Three	XIII	Thirteen	L	Fifty	DC	Six hundred
IV	Four	XIV	Fourteen	LX	Sixty	DCC	Seven hundred
V	Five	XV	Fifteen	LXX	Seventy	DCCC	Eight hundred
VI	Six	XVI	Sixteen	LXXX	Eighty	CM	Nine hundred
VII	Seven	XVII	Seventeen	XC	Ninety	M	Thousand
VIII	Eight	XVIII	Eightteen	C	Hundred		
IX	Nine	XIX	Nineteen	CC	Two hundred		
X	Ten	XX	Twenty	CCC	Three hundred		

Retrieved from: www.englishstudyhere.com

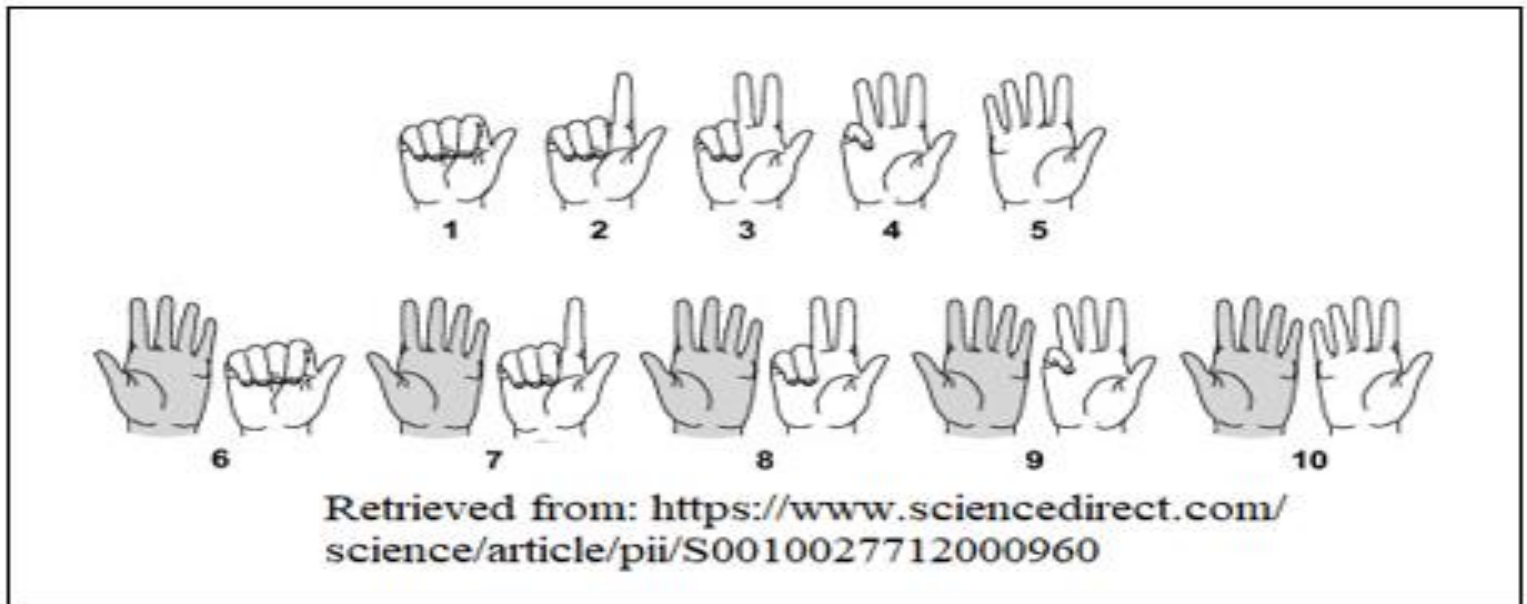
Brief History of Number System

2.1.3 Decimal System

- A decimal (or denary) system is a numeral system that has the number ten as its base. The term decimal is likewise utilized for a number written in this system, or for a division expressed utilizing this system.
- A number written in decimal representation includes the utilization of at least one of ten distinctive system or fundamental units, called digits. The digits are regularly utilized with a decimal separator, which shows the beginning of a fractional part. The decimal separator might be a dab, a period, or a comma.

Brief History of Number System

- The number ten is the count of the total number of fingers and thumbs on a person's two hands (or toes on two feet). In many languages, the word *digit* or its translation is also the anatomical term referring to fingers and toes. In English, the term *decimal* (from Latin *decimus*) means "tenth," *decimate* means "reduce by a tenth," and *denary* (Latin *denarius*) means "the unit of ten."



Brief History of Number System

2.1.4 Rounding off-number

Rounding means making a number easier but keeping its value near what it was. The outcome is less precise, however simpler to utilize

Here's the general guideline for adjusting:

- If the number are rounding is followed by 5, 6, 7, 8, or 9, round the number up. Example: 38 rounded to the closest ten is 40
- If the number are rounding is followed by 0, 1, 2, 3, or 4, round the number down. Example: 33 adjusted to the closest ten is 30

Brief History of Number System

2	6	2	4	3	4	5	.	2	3	4	5
millions	hundred thousands	ten thousands	thousands	hundreds	tens	ones	decimal point	tenths	hundredths	thousandths	ten thousandths

Retrieved from: https://www.varsitytutors.com/hotmath/hotmath_help/topics/rounding-numbers

Brief History of Number System

2.1.5 Engineering Notation

- Engineering notation is an interpretation of scientific representation in which the exponent in expression of the form is selected to always be divisible by 3. Numbers of forms, for example, 340, and therefore compare to engineering notation, while numbers, for example, and do not.

PREFIX	SYMBOL	MULTIPLIER	EXPONENT FORM
exa	E	1, 000, 000, 000, 000, 000, 000	10^{18}
peta	P	1, 000, 000, 000, 000, 000	10^{15}
tera	T	1, 000, 000, 000, 000	10^{12}
giga	G	1, 000, 000, 000	10^9
mega	M	1, 000, 000	10^6
kilo	k	1, 000	10^3
hecto	h	100	10^2
deca	da	10	10^1
Basic Unit	Basic Unit	1	10^0
deci	d	0.1	10^{-1}
centi	c	0.01	10^{-2}
milli	m	0.001	10^{-3}
micro	μ	0.000,001	10^{-6}
nano	n	0.000,000,001	10^{-9}
pico	p	0.000,000,000,001	10^{-12}
femto	f	0.000,000,000,000,001	10^{-15}
atto	a	0.000,000,000,000,000,001	10^{-18}

Brief History of Number System

2.1.6 Scientific Notation

- Scientific Notation (also referred to as scientific form or standard record frame, or standard frame in the UK) is a method for communicating numbers that are too large or too small to be usefully written in decimal form. It is normally utilized by scientist, mathematicians and engineer, to some extent since it can simplify a certain arithmetic operation. On scientific calculators it is usually known as "SCI" display mode.

Exponent 10^n	Metric Prefix	Symbol
12	tera	T
9	giga	G
6	mega	M
3	kilo	k
-3	milli	m
-6	micro	μ
-9	nano	n
-12	pico	p

Retrieved from: <http://nolancardona0783.weebly.com/scientific-and-engineering-notation.html>

Brief History of Number System

2.2 A BRIEF HISTORY OF BINARY NUMBER SYSTEM

- The simplest of all positional number system is the binary number system. It has a base-or radix of the binary system is 2, meaning it has only two digits represented by 0 and 1 which appear in a binary representation of any number. The two digits, 0 and 1, are considered as the two states (off/on) and these states are used to carry instructions and store data in computers. In general, this element represents just one bit which is referred to as binary digit. Thus, the binary system underlies modern technology of electronic digital computers.

Brief History of Number System

- The first electronic digital computer was built at the University of Pennsylvania and called the Electronic Numerical Integrator and Calculator (ENIAC), the first electronic computer was invented seventy-one years ago. The invention of the binary system dated almost three centuries back.
- Gottfried Wilhelm Leibniz (1646-1716), the co-inventor of Calculus, published a paper about his invention in 1701 in a paper essay D'une Nouvelle Science Des Nombres about his invention. The paper was submitted to the Paris academy to mark his election to the academy. Moreover, it took another twenty years for the discovery to happen just like it took a few hundred years to develop a binary convertor.

Brief History of Number System

2.2.1 Understanding Binary Numbers

- The formal beginnings of the binary system, is around 200 BC, Pingala, an Indian writer, introduced sophisticated mathematical concepts that described metrics and gave the world its first ever description of a binary number system.

Brief History of Number System

2.2.2 Applications of Number system

- The most common application for the number system in computer technology. Wherein a two-digit number system used in digital encoding is what all computer language and programming are based on. Using the data and then depicting it with restrained bits of information is what makes up the digital encoding process. The controlled information comprises of the binary system's 0s and 1s. An example of this is the images on your computer screen. A binary line of each pixel is used to encode these images.

Brief History of Number System

2.2.3 Advantage of the Binary Number System

- The binary number system is suitable for a number of things. For example, to add numbers, a computer flips switches. By adding binary numbers to the system, you can simulate computer is performing addition. Also, there are two main reasons to use this number system for computers. First is that it can provide a safety range for reliability. Secondary and most importantly, it helps minimize the circuitry needed in electronic design and lower the space required, and the energy consumed and costs that are spent.

Brief History of Number System

2.3 GENERATION OF COMPUTER

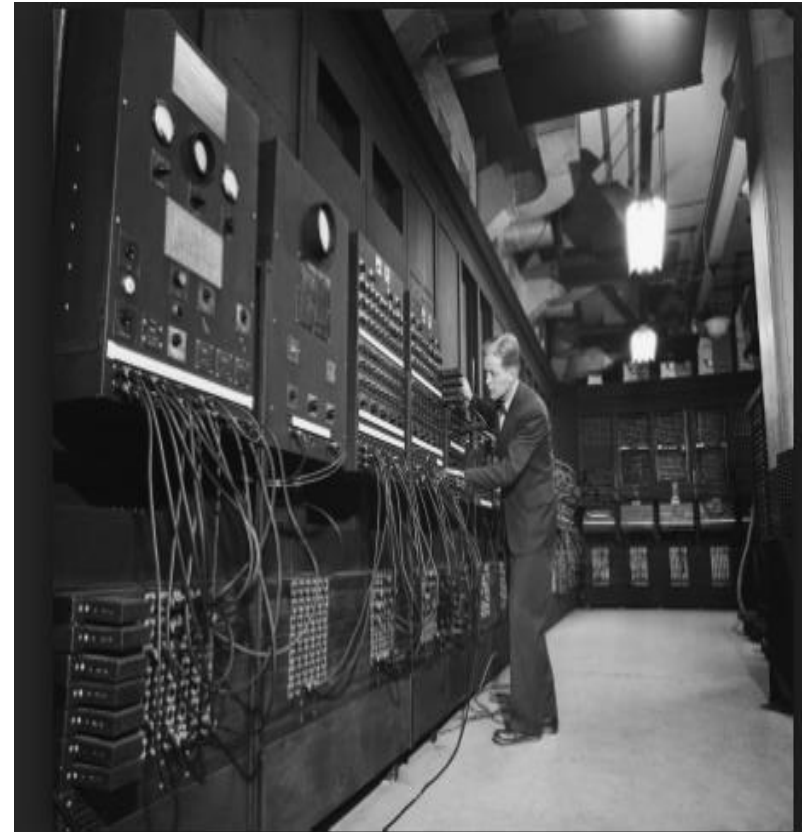
The evolution of computer took place, can be divided into five distinct phases, basis of the type of switching circuits known as Generations of Computers.

1. First Generation Computers –1942 to 1954
2. Second Generation of Computers -1955 to 1964
3. Third Generation of Computers- 1965 to 1974
4. Fourth Generation Computers- 1975 – till now
5. Fifth Generation of Computers – Still in Process

Brief History of Number System

2.3.1 First Generation Computers (1950's)

Electronic computers were the limited possessions of scientists, engineers, and the military until 1951. No one had tried to create an electronic digital computer for business. Eckert and Mauchly, the first to try the digital computer. When the University of Pennsylvania learned of them plans to transform ENIAC into a commercial product, University officials stated that the university owned the duo's patent.



Brief History of Number System



- These early computers used vacuum tubes as circuitry and magnetic drums for memory. The problem with vacuum tubes was that they failed frequently, costing a fortune to run, made of inefficient materials which generated a lot of heat, and sucked huge electricity and subsequently generated a lot of heat which caused ongoing breakdowns making the first-generation computers were down (not working) ample of the time. The advantage of vacuum tubes technology is that it made the advent of Electronic digital computer. Vacuum tubes were only electronic devices available during those days which made computing possible.

Brief History of Number System

A. Computer Characteristics & Capabilities.

- Cost – cost was very high.
- Language – Machine and Assembly Language.
- Power – high power Consumption and it generated much heat.
- Reliability – high failure rate, Failure of circuits per second.
- Size – Relatively big size. Size was equivalent to a room.
- Speed – slow speed, hundred instructions per second.

Brief History of Number System

B. Trends and Developments in Computer Hardware.

- Input Media – Punched cards & paper tape
- Main Component – based on vacuum tubes
- Main memory –Magnetic drum
- Output Media – Punched card & printed reports.
- secondary Memory – Magnetic drum & magnetic tape.

Brief History of Number System

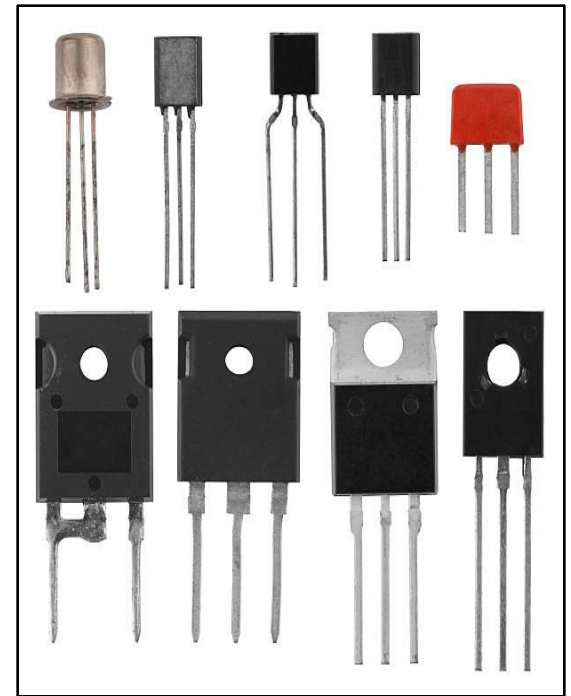
Examples

- ENIAC
- UNIVAC
- Mark –I
- Mark-III
- IBM 700 series
- IBM 700 series
- IBM 701 series
- IBM 709 series
- etc.

Brief History of Number System

2.3.2 Second Generation Computers (Early 1960's)

Second-generation computers were created with transistors instead of vacuum tubes, these computers were faster, smaller, and more reliable than first-generation computers. The replacement of vacuum tubes by transistors saw the advent of the second generation of computing. Although first invented in 1947 by Bell Laboratories, transistors weren't used significantly in computers until the end of the 1950s.



Brief History of Number System

A. Computer Characteristics & Capabilities:

- Cost – cost Slightly lower than first generation.
- Language – Assembly Language and High-level languages like FORTRAN, COBOL, BASIC.
- Power– Low power Consumption.
- Reliability – Failure of circuits per days.
- Size – Smaller than first generation Computers.
- Speed – Relatively fast as compared to first generation, thousand instructions per second.

Brief History of Number System

B. Trends and Developments in Computer Hardware:

- Input Media – Punched cards
- Main Component – Based on Transistor.
- Main Memory – Magnetic core.
- Output Media – Punched card & printed reports.
- Secondary Memory – Magnetic tape & magnetic Disk.

Brief History of Number System

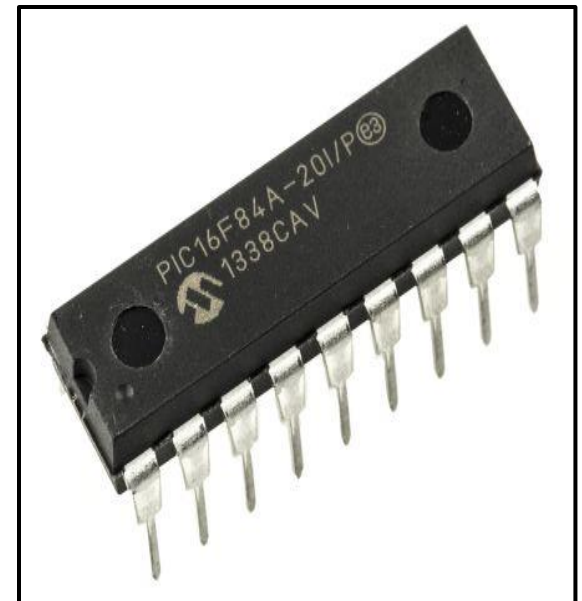
Example

- IBM-7000
- CDC 3000 series
- PDP1
- PDP3
- PDP 5
- PDP8
- ATLAS
- IBM-7094
- etc.

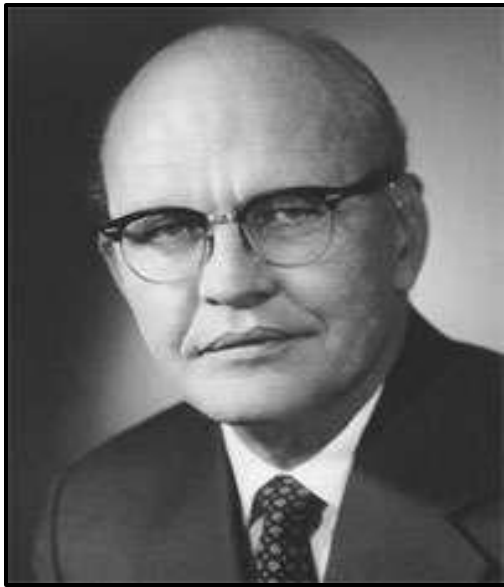
Brief History of Number System

2.3.3 Third Generation Computers (Mid-1960s to Mid-1970s)

The third-generation computers were introduced in 1964. They used Integrated Circuits (ICs). These ICs are popularly known as Chips (called semiconductors). A single IC has many transistors, registers and capacitors built on a single thin slice of silicon. In this phase, transistors were now being miniaturized led to a massive increase in speed and efficiency of these machines.



Brief History of Number System



JACK ST. CLAIR KIRBY

- Invented by Jack St. Clair Kirby and Robert Noyce in 1958, integrated circuits assured to cut the cost of computer production significantly because ICs could duplicate the functions of transistors at a small fraction of a transistor's cost. The earliest ICs, using a technology now called small-scale integration (SSI), could contain up to 10 to 20 transistors on a chip. By the late 1960s, engineers had achieved medium-scale integration (MSI), which placed between 20 and 200 transistors on a chip. In the early 1970s, large-scale integration (LSI) was achieved, in which a single chip could hold up to 5,000 transistors.

Brief History of Number System

- Scientists knew that more powerful computers could be built by creating more complex circuits. But because these circuits had to be wired by hand, these computers were too complex and expensive to build. With integrated circuits, new and innovative designs became possible for the first time. With ICs on the scene, it was possible to make smaller, inexpensive computers that more organizations could afford to acquire



ROBERT NOYCE

Brief History of Number System

A. Computer Characteristics & Capabilities:

- Cost – cost lower than Second generation.
- Language– High level languages like PASCAL, COBOL, BASIC, C etc.
- Power– Low power Consumption.
- Reliability – Failure of circuits in Weeks.
- Size – Smaller than Second Generation Computers. Disk size mini computers.
- Speed – Relatively fast as compared to second generation, Million instructions per second (MIPS).

Brief History of Number System

B. Trends and Developments in Computer Hardware:

- Main Component – Based on Integrated Circuits (IC)
- Primary Memory – Magnetic core.
- Secondary Memory– Magnetic Tape & magnetic disk.
- Input Media – Key to tape & key to disk
- Output Media – Printed reports & Video displays.

Brief History of Number System

Example

- IBM-307 Series
- CDC 7600 series
- PDP (Personal Data processor) II
- etc.

Brief History of Number System

- **2.3.3.1 Moore's Law**

- Moore's law is the perception that over the historical background of processing equipment, the quantity of transistors on integrated circuits doubles approximately every two years. The period regularly cited as "year and a half" is because of Intel official David House, who anticipated that period for a multiplying in chip execution (being a blend of the impact of something beyond transistors and their being quicker).



<https://www.wired.com/2015/04/how-gordon-moore-made-moores-law/>

Brief History of Number System

Figure 1 shows the transistor counts from 1971-2011, the graph shows the exponential relationship between the date of introduction and transistor count from 1971 to 2011. Based on the graph as the time increased the amount of transistor count double the number of transistor. Moore's law predicts that this trend will continue into the foreseeable future

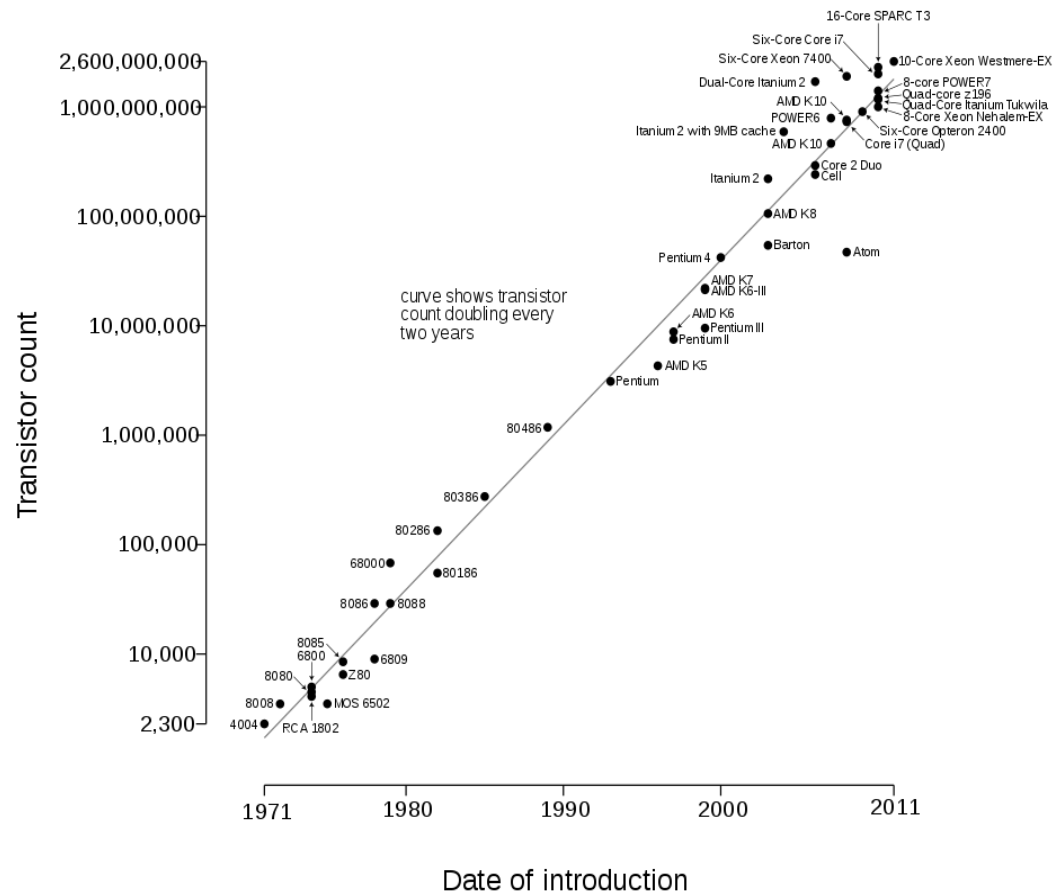


Figure 1. Microprocessor transistor counts 1971-2011 & Moore's Law

Brief History of Number System

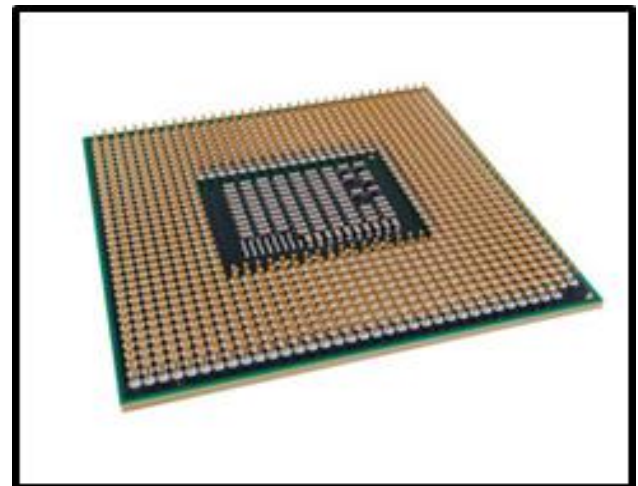
Table 2. Scale of Integration of Integrated Circuit

Type	No. of transistors	Typical Applications
Small Scale Integration (SSI)	1-100	Logic gates, op-amps, linear applications.
Medium Scale Integration (MSI)	101-1,000	Registers, filters, and so on
Large Scale Integration (LSI)	1,001-100,000	8-bit microprocessors, up to 64 kbit ROMs and RAMs, Analogue-to-Digital converters, and so on
Very Large Scale Integration (VLSI)	100,001-500, 000	16/32-bit microprocessors, up to 256 kbit ROMs/RAMs, signal processors.
Ultra High Scale Integration (UHSI)	>500,001	64-bit microprocessors, 8 Mbit RAMs, real-time and image processors.
Gigantic Scale Integration (GSI)	>10,000,000	64 Mbit RAMs, integrated multi-processors.

Brief History of Number System

2.3.4 The Fourth Generation (1975 to the Present)

- In the early 1970s, Dr. Ted Hoff, an Intel Corporation engineer, was given the task of designing an integrated circuit to power a digital watch. Beforehand, these circuits had to be restructured every time a new model of the watch emerge. Dr. Hoff decided that he could avoid costly redesigns by producing a tiny computer on a chip and was called as Intel 4004, and considered the world's first **microprocessor**.



Brief History of Number System

Steve Jobs and Steve Wozniak considered to be the young entrepreneurs at that time dreamed of creating an “appliance computer.” They wanted a simple microcomputer that could take it out of the box, plug it in, and use it (plug and play). Jobs and Wozniak sell a Volkswagen for \$1,300 to raise a capital to setup a shop in a garage. As they founded the Apple Computer, Inc, in April 1977. The first product was a processor board intended for the hobbyist, since the company gained an experience in building Apple1 they developed Apple II computer system.



Steve Jobs and Steve Wozniak

Brief History of Number System

A. Computer Characteristics & Capabilities:

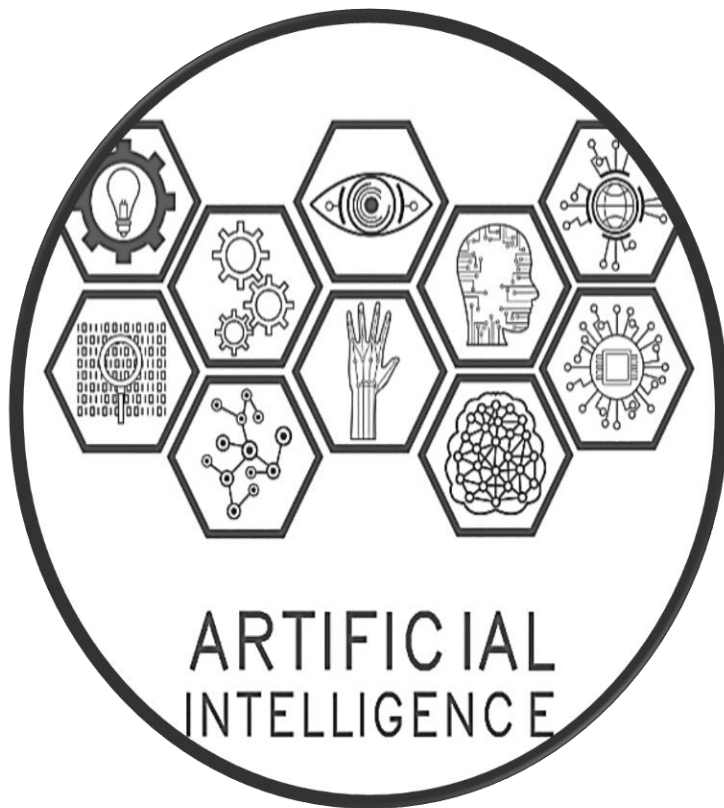
- Cost – Cost lower than third generation.
- Language– High level languages like C++, KL1, RPG, SQL.
- Power– Low power Consumption.
- Reliability – Failure of circuits in months.
- Size – Typewriter size micro Computer.
- Speed – Relatively fast as compared to Third generation, Tens of Millions of instructions per second.

Brief History of Number System

B. Trends and Developments in Computer Hardware:

- Input Media – keyboard.
- Main Component – Large scale integrated (LSI) Semiconductor circuits called MICRO PROCESSOR or chip and VLSI (Very Large scale integrated).
- Main Memory – Semi-conductor memory like RAM, ROM and cache memory is used as a primary memory.
- Output Media – Video displays, Audio responses and printed reports.
- Secondary Memory – Magnetic disk, Floppy disk, and Optical disk (CD, DVD).
 - Example – CRAY 2, IBM 3090/600 Series, IBM AS/400/B60 etc.

Brief History of Number System



- In this generation of computer Artificial Intelligence (AI) concept is adopted. The computers have intelligence quality, default assumptions, Decision making capability etc. through these concepts expert systems, Knowledge based systems, Decision Support System are developed. Robots are the common example of this type of system.

THANK YOU FOR LISTENING! 😊